MATH_340_Fall_2022 - 2022-12-11T171817.455

Sunday, December 11, 2022 5:18 PM

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Math 340: Elementary Matrix and Linear Algebra

FINAL EXAM

Friday, December 16, 2022

Please circle your instructor and your Discussion Session (start time given) in the table below.

Dr. Skylar Grey	Dr. Boya Wen	Dr. Ziquan Yang
Zaidan Wu, T 3:30	Alex Pahlow, M 12:05	Connor Simpson, T 7:45
Zaidan Wu, T 4:35	Ugur Cin, M 12:05	Asvin Gothandaraman, T 8:50
Lauren Neurdorf, R 3:30	Ugur Cin, W 12:05	Connor Simpson, R 7:45
Hyun Jong Kim, R 9:55	Pubo Huang, W 1:20	Asvin Gothandaraman, R 9:55
Hyun Jong Kim, T 9:55	Woojin Kim, M 3:30	
Hyun Jong Kim, R 8:50	Woojin Kim, W 3:30	
	Ugur Cin, W 4:35	
	Woojin Kim, M 2:25	

Dr. Kaitlyn Phillipson	Dr. Zane Li
Emma Thomas, R 11:00	Jacob Denson, M 1:20
Emma Thomas, R 12:05	Jacob Denson, M 2:25
Emma Thomas, R 1:20	Jacob Denson, W 2:25

Name: Key ______________________ Wisc email: ______________________

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: ______________________________

READ THE FOLLOWING INFORMATION. **DO NOT BEGIN THIS EXAM UNTIL SIGNALED TO DO SO.**

- This is a 120-minute exam. It consists of 12 problems for a total of 100 points; the exam is 9 sheets of paper, including this cover sheet. It is your responsibility to make sure that you have a complete exam.
- Books, notes, calculators, and other aids are not allowed.
- Problems are spaced out to allow ample room for work. You may use the last page for scratch work, but it will not be graded. Do not unstaple or remove pages as they can be lost in the grading process. **An incomplete exam will result in an automatic zero.**
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.
- Notation reminders: M_{mn} is the vector space of $m \times n$ real matrices, P_d is the vector space of polynomials of degree at most d, $\mathbf{0}$ denotes the zero vector in a vector space V. $\mathbb{R}^n$ uses the normal vector addition/scalar multiplication unless otherwise noted.

1. (12 points, 2 point each) Are each of the following true or false statements? Circle your answer. Justification is not necessary.

a.) **T** **F** If $\lambda = 0$ is an eigenvalue for A, then A is not invertible.

b.) **T** **F** Any set of five vectors in $\mathbb{R}^4$ will span $\mathbb{R}^4$.

c.) **T** **F** If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set of vectors, then so is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 - \mathbf{v}_1\}$.

d.) **T** **F** If A and B are similar, then $\text{rank}(A) = \text{rank}(B)$.

e.) **T** **F** All diagonalizable matrices are also invertible.

f.) **T** **F** If $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^3$, then S is linearly independent.

2. (7 points) Consider the vector space M_{22} with normal matrix addition and scalar multiplication. Consider the subset W of M_{22} of matrices which are NOT invertible. Show through explicit counterexample(s) that this set is NOT a subspace of M_{22}. Fully justify your answer.

let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \rightsquigarrow \det(A) = 1 \cdot 0 - 0 \, 0 = 0 \Rightarrow A$ is not invertible.

let $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \det(B) = 0 \cdot 1 - 0 \cdot 0 = 0 \Rightarrow B$ is not invertible.

$A+B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightsquigarrow \det(A+B) = 1 \cdot 1 - 0 \, 0 = 1 \Rightarrow A+B$ is invertible!

so it's not closed under $+$, so W is not a subspace of M_{22}.

3. (6 points total)

a. (3 points total) Consider the following vector spaces. Give the dimension of each (no justification is necessary).

1. M_{22}
   Dimension: 4
2. P_2
   Dimension: 3
3. W, where W is the subspace of $\mathbb{R}^4$ spanned by

$$\left\{ \underbrace{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}3\\2\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix}}_{\text{lin. indep.}}, \overbrace{\begin{bmatrix}4\\2\\0\\1\end{bmatrix}}^{\text{sum of prev. 3.}} \right\}$$

   Dimension: 3

b. (3 points) Of the vector spaces in part (a), which of them (if any) are isomorphic to $\mathbb{R}^3$? Briefly (1 sentence) justify your answer.

P_2 and W, b/c they have the same dim. as $\mathbb{R}^3$ ($\dim(\mathbb{R}^3) = 3$)

4. (10 points total) (This problem has overflow room on next page if needed)

Consider the set of vectors in $\mathbb{R}^3$:

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\}$$

a. (4 points) Verify that S is a basis for $\mathbb{R}^3$. Briefly explain your work.

$$\begin{array}{ccc} + & - & + \end{array}$$
$$\det \begin{bmatrix} 1 & 4 & 0 \\ 1 & 2 & 2 \\ 1 & 1 & 0 \end{bmatrix} = -2(1\cdot 1-4) = 6 \neq 0 \rightsquigarrow \text{lin indep}$$

and since there's $\underline{3}$ of them, this must form a basis in $\mathbb{R}^3$.

(can also argue via row reduction)

b. (6 points) Suppose $L : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation with

$$L\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad L\left(\begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}, \quad L\left(\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}$$

Using the fact that S is a basis, find $L\left(\begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}\right)$. Show all work.

Solve:

$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & -1 \\ 1 & 2 & 2 & 3 \\ 1 & 1 & 0 & -1 \end{array}\right] \xrightarrow[-r_1+r_3\to r_3]{-r_1+r_2\to r_2} \left[\begin{array}{ccc|c} 1 & 4 & 0 & -1 \\ 0 & -2 & 2 & 4 \\ 0 & -3 & 0 & 0 \end{array}\right]$$

$$\Big\downarrow \; -\tfrac{1}{3}r_3\to r_3,\ r_2 \leftrightarrow r_3$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & -2 & 2 & 4 \end{array}\right]$$

$$\Big\downarrow \; \begin{array}{l} 2r_2+r_3\to r_3 \\ -4r_2+r_1\to r_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 4 \end{array}\right] \quad \begin{array}{l} a=-1 \\ b=0 \\ c=2 \end{array}$$

(more room for Problem 4, if needed):

$$L\left(\begin{bmatrix}-1\\3\\-1\end{bmatrix}\right)=L\left(-1\begin{bmatrix}1\\1\\1\end{bmatrix}+2\begin{bmatrix}0\\2\\0\end{bmatrix}\right)=$$

$$=-1L\left(\begin{bmatrix}1\\1\\1\end{bmatrix}\right)+2L\left(\begin{bmatrix}0\\2\\0\end{bmatrix}\right)=\begin{bmatrix}-2\\0\\-2\end{bmatrix}+2\begin{bmatrix}-1\\1\\3\end{bmatrix}$$

$$=\boxed{\begin{bmatrix}-4\\2\\4\end{bmatrix}}$$

5. (7 points) Suppose $A = \begin{bmatrix} c & 1 \\ d & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ -1 & -2 \end{bmatrix}$. For which value(s) of c and d is AB both **invertible** and **symmetric**? Show all work.

$$AB = \begin{bmatrix} c & 1 \\ d & 2 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 3c-1 & -2 \\ 3d-2 & -4 \end{bmatrix}$$

$=$ sym.

$$3d-2=-2$$
$$d=0$$

$$\det\begin{bmatrix} 3c-1 & -2 \\ -2 & -4 \end{bmatrix} = (3c-1)(-4)-4=0$$

$$3c-1 = \frac{+4}{-4} = -1$$

$$3c=0$$

$c=0 \leftarrow$ not invertible

$$\boxed{\text{Need } c \neq 0 \text{ and } d=0}$$

6. (8 points total) (This problem has overflow room on next page if needed) Suppose A is a 5×5 matrix with the following reduced row echelon form:

$$RREF(A) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} . \begin{bmatrix} 2 \\ -2 \\ 4 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Again, this is the reduced row echelon form of A. Answer each question with a brief explanation.

a.) (2 points) What is the nullity of A? Explain your answer. If there is not enough information, explain.

1, b/c there's one free variable.

b.) (2 points) What is the determinant of A? Explain your answer. If there is not enough information, explain.

$\det(A) = 0 \rightarrow$ its reduced eclelon form has a row of zeros, so it doesn't reduce to I_5.

c.) (2 points) Is $\begin{bmatrix} 2 \\ -2 \\ 4 \\ -2 \\ 0 \end{bmatrix} = \vec{x}$ a solution to $A\mathbf{x} = \mathbf{0}$? Explain your answer. If there is not enough information, explain.

Yes, b/c $RREF(A)\vec{x} \stackrel{?}{=} \vec{0}$.

d.) (2 points) Is $\lambda = 1$ an eigenvalue of A? Explain your answer. If there is not enough information, explain.

Can't tell $\rightarrow$ reducing can change e-values.

(Overflow room for Problem 6, if needed):

7. (6 points total) Consider $\mathbb{R}^3$ with the standard inner product (also known as the dot product), and let $\mathbf{u} = \begin{bmatrix} 1/2 \\ 0 \\ a \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -4 \\ b \\ 1 \end{bmatrix}$.

a. (3 points) For which value(s) of a (if any) is $\mathbf{u}$ a unit vector? Show all work.

$$\|\vec{u}\| = \sqrt{(1/2)^2 + a^2} = 1$$

$$a^2 = 1 - 1/4 = 3/4$$

$$a = \pm\sqrt{\frac{3}{4}}$$

b. (3 points) For which value(s) of a and b (if any) are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? Show all work.

$$\vec{u}\cdot\vec{v} = 1/2(-4) + 0\cdot b + a\cdot 1 = -2 + a = 0$$

$$\boxed{\begin{array}{l} a=2 \\ b \text{ can be anything} \end{array}}$$

8. (8 points total) Consider $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ in $\mathbb{R}^2$, and define the function

$$(\mathbf{u}, \mathbf{v}) = u_2 v_2 - u_2 v_1 - u_1 v_2 + 2u_1 v_1.$$

a. (4 points) Verify the following inner product property for $(\mathbf{u}, \mathbf{v})$:
$(c\mathbf{u}, \mathbf{v}) = c(\mathbf{u}, \mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^2$ and scalar c.

$$(c\vec{u}, \vec{v}) = \left(\begin{bmatrix} cu_1 \\ cu_2 \end{bmatrix}, \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right) = cu_2v_2 - cu_2v_1 - cu_1v_2 + 2(cu_1)v_1$$

$$= c(u_2v_2 - u_2v_1 - u_1v_2 + 2u_1v_1)$$

$$= c(\vec{u}, \vec{v})$$

b. (4 points) Assume the remaining inner product properties are true. Using this inner product $(\mathbf{u}, \mathbf{v})$, find a unit vector in the same direction as $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

$$\|\vec{u}\| = \sqrt{(\vec{u}, \vec{u})} = \sqrt{2 \cdot 2 - 2 \cdot 1 - 1 \cdot 2 + 2 \cdot 1 \cdot 1}$$

$$= \sqrt{4 - 2 - 2 + 2} = \sqrt{2}.$$

$\boxed{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}}$ is a unit vector.

9. (7 points total) Consider $A = \begin{bmatrix} -6 & -4 \\ 8 & 6 \end{bmatrix}$.

a. (5 points) Find the eigenvalues of A.

$$\det(\lambda I - A) = \det\begin{bmatrix} \lambda+6 & 4 \\ -8 & \lambda-6 \end{bmatrix} = (\lambda+6)(\lambda-6) + 32$$

$$= \lambda^2 - 36 + 32$$

$$= \lambda^2 - 4 = (\lambda+2)(\lambda-2)$$

$$\boxed{\lambda = 2, -2}$$

b. (2 points) Is A diagonalizable? (Hint: you do **not** need to calculate the eigenvectors to solve this problem). Briefly justify your answer.

Yes! b/c A is 2×2 w/ 2 distinct e-values.

10. (8 points) For $A = \begin{bmatrix} 5 & 4 & 2 \\ -1 & 1 & -1 \\ -1 & -2 & 2 \end{bmatrix}$, find a basis for the eigenspace associated to the eigenvalue $\lambda = 3$. Show all work.

$$(\lambda I - A) = \begin{bmatrix} \lambda-5 & -4 & -2 \\ 1 & \lambda-1 & 1 \\ 1 & 2 & \lambda-2 \end{bmatrix} \xrightarrow{\lambda=3} \begin{bmatrix} -2 & -4 & -2 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\Big\downarrow \; -\tfrac{1}{2}r_1 \to r_1,\ -2r_1 + r_2 \to r_2,\ -2r_1 + r_3 \to r_3$$

$$\begin{array}{ccc} a & b & c \end{array}$$
$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$b = r,\ c = s.$

$a = -2r - s.$

$$\begin{bmatrix} -2r-s \\ r \\ s \end{bmatrix} = r\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + s\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Basis: $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

11. (12 points total) (This problem has overflow room on next page if needed)

Consider $L: M_{22} \to \mathbb{R}^3$ by $L\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a \\ b+c \\ b-d \end{bmatrix}$.

a.) (5 points) Find a basis for the kernel of L. Show all work.

want: $L\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} a \\ b+c \\ b-d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow a=0,\ b+c=0,\ b-d=0.$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \xrightarrow{-r_2+r_3 \to r_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 \end{bmatrix} \xrightarrow[-r_3 \to r_3]{r_3+r_2 \to r_2} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

$a=0$

$b-d=0 \rightarrow b=r$

$c+d=0 \quad c=-r$

$d=r$

elt. of M_{22}

$$\begin{bmatrix} 0 & r \\ -r & r \end{bmatrix}$$

$$\boxed{\text{Basis: } \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}}$$

b.) (2 points) Using your answer for (a), what is the dimension of the kernel (also known as nullity of L)? Fully justify your answer.

nullity $= 1 \rightarrow$ b/c kernel has 1 basis element.

c.) (3 points) Find $\dim \operatorname{range}(L)$. Fully justify your answer.

$$\text{nullity} + \dim \operatorname{range} L = \underset{\dim(M_{22})}{\overset{}{\underset{\parallel}{4}}} \rightarrow \dim \operatorname{range} L = 4 - 1 = 3.$$

d.) (2 point) Using your answer from (c), is L onto (also known as surjective)? Fully justify your answer.

Yes, b/c range is 3-dimensional, matching $\mathbb{R}^3$.

(overflow room for Problem 11, if needed):

12. (9 points total) Define $L : P_2 \to P_2$ by

$$L(at^2+bt+c) = (a-b)t^2 + 2ct + (a-b+c).$$

You may use without proof that L is a linear transformation.

a.) (5 points) Using the ordered basis $S = (1, t^2, t)$ (**Note the order!!**) for P_2, write the matrix A representing L. Show all work.

$L(1) = 2 \cdot 1 t + 1 = 2t+1 \rightsquigarrow \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$

$L(t^2) = (1-0)t^2 + 2 0 \cdot t + (1-0+c) = t^2+1 \rightsquigarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$L(t) = -1t^2 + 2 \cdot 0 \cdot t + (0-1+0) = -t^2 - 1 = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 2 & 0 & 0 \end{bmatrix}$$

b.) (4 points) Using your answer to part (a): Is L invertible? Fully justify your answer.

$\det(A) = 2(1(-1) - (1(-1))) = 2(-1+1) = 0.$

No, A isn't invertible, so L isn't invertible.

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